

[1] Device	Purpose	Assume	SS EE
Nozzle	speed up flow $+$	$\dot{W} = \dot{Q} = \Delta p_e = 0$ $+$	$\Delta h + \Delta ke = 0$ $+$
Diffuser	slow down flow $+$	$\dot{W} = \dot{Q} = \Delta p_e = 0$ $+$	$\Delta h + \Delta ke = 0$ $+$
Throttle	control flow by dropping pressure $+$	$\dot{W} = \dot{Q} = \Delta ke = \Delta p_e = 0$ $+$	$\Delta h = 0$ $+$
Heat Exchanger	Exchange heat, sometimes from one fluid to another $+$	$\dot{W} = \Delta ke = \Delta p_e = 0$ $+$	$\dot{m} \Delta h = \dot{Q}$ $+$
Compressor	Increase enthalpy of fluid by increasing pressure, temperature $+$	$\dot{Q} = \Delta ke = \Delta p_e = 0$ $+$	$\dot{m} \Delta h = -\dot{W}$ $+$ $\downarrow$ $+$ for sign
Turbine	Extract enthalpy from fluid as pressure drops $+$	$\dot{Q} = \Delta ke = \Delta p_e = 0$ $+$	$\dot{m} \Delta h = -\dot{W}$ $+$ $\downarrow$ $+$ for sign

[2] Given: $\dot{m} = 50 \text{ kg/s}$ $\dot{W} = -2.7457 \text{ MW}$
 $T_1 = 300 \text{ K}$ $p_2 = 1.7674 \text{ kg/m}^3$
 $u_1 = 170 \text{ m/s}$ $A_3/A_2 = 3$ $u_3 = 300 \text{ m/s}$

Assume: air = ideal gas $+$ $1/2$ steady flow $+$ $1/2$
 $c_p = \text{const}$ $+$ $1/2$ fan $\rightarrow \dot{Q} = \Delta ke = \Delta p_e = 0$ $+$ $1/2$
 quasi-steady $+$ $1/2$ nozzle $\rightarrow \dot{W} = \dot{Q} = \Delta p_e = 0$ $+$ $1/2$

a) $\dot{m} (\Delta h + \Delta ke + \Delta p_e) = \dot{Q} - \dot{W} + 2$

fan $\rightarrow \dot{m} \Delta h = -\dot{W} + 2$

ideal gas / $c_p = \text{const}$ $\rightarrow \dot{m} c_p (T_2 - T_1) = -\dot{W} + 2$

$$T_2 = T_1 - \frac{\dot{W}}{\dot{m} c_p} = 300 - \frac{(-2.7457 \times 10^6)}{(50)(1001)}$$

$$\boxed{T_2 = 354.86 \text{ K}} \quad +1$$

b) steady flow: $\dot{m}_2 = \dot{m}_3 \Rightarrow \rho_2 u_2 A_2 = \rho_3 u_3 A_3 \rightarrow \rho_3 = \frac{\rho_2 u_2 A_2}{u_3 A_3} + 2$

$\Delta ke = 0$ for fan $\rightarrow u_2 = u_1 = 170 \text{ m/s} \rightarrow \rho_3 = \frac{(1.7674)(170)}{(300)(3)} \rightarrow \rho_3 = 0.334 \frac{\text{kg}}{\text{m}^3} \quad +1$

→ nozzle : $\Delta h + \Delta ke = 0$ +2

→ ideal gas : $C_p(T_3 - T_2) + \frac{1}{2}(v_3^2 - v_2^2) = 0$ +2
 $C_p = \text{const}$

$$T_3 = T_2 - \frac{1}{2C_p}(v_3^2 - v_2^2)$$

$$= 354.86 - \frac{1}{2002}(300^2 - 170^2)$$

$$\boxed{T_3 = 324.3 \text{ K}} \quad +1$$

[3] Given: $\dot{m} = 10 \text{ kg/s}$ Find: $\dot{W} = ?$ Assume: S.C.S. +1/2 +1

+2 $P_1 = 10 \text{ MPa}$

$T_1 = 900 \text{ K}$

$x_2 = 1$

$P_2 = 0.1 \text{ MPa}$

: quasi-st. +1/2 +1

: steady-flow +1/2 +1

: $\dot{Q} = \Delta ke = \Delta pe = 0$ +1/2 +1
 for turbine

→ steady flow: $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$ +2

→ turbine: $\dot{m} \Delta h = -\dot{W}$ +2

$\dot{W} = -\dot{m}(h_2 - h_1) \rightarrow \text{State 1: } P_1 = 10 \text{ MPa (Table D.2)} \quad +1$

$= -10(2674.9 - 3691.6)$

$T_1 = 900 \text{ K}, T_1 > T_{\text{sat}} \quad +1$

$\Rightarrow \text{Vapor (Table D.3P)}$

$h_1 = 3691.6 \text{ kJ/kg} \quad +2$

$$\boxed{\dot{W} = 10.167 \text{ MW}} \quad +2$$

↑
 +1 for
 correct
 units

→ State 2: $P_2 = 0.1 \text{ MPa}$

$x_2 = 1 \text{ (Table D.2)} \quad +1$

$h_2 = h_g = 2674.9 \text{ kJ/kg} \quad +2$

[4] Statement 1: you can't devise a power cycle where heat is completely converted to work +2 → some heat has to be rejected +1

$\Rightarrow$ Implication: $\eta_{\text{th}} < 1$ +2, always some energy loss in conversion of heat to work, even in a frictionless world. Will never use all the energy in the fuel. +1

Statement 2: processes have a natural directionality and if you want to move a system in the opposite direction of its spontaneous direction, it requires work input. +2

→ Implication: a cycle will always include some non-spontaneous processes, so you're always going to have to do work. +1

Statement 3: if any losses/irreversibilities are present, entropy will be generated. +2

→ In a cycle, this increases the entropy of the surroundings. +1

Statement 4: spontaneous processes move systems towards equilibrium, where entropy is maximized. +2

→ Implication: losses and irreversibilities are inescapable. +1

[5] $\eta_{th} = \text{work you got out} / \text{heat you put in}$ +5

⇒ $\dot{W} = \eta_{th} \dot{Q}_{in}$, where $\dot{Q}_{in}$ comes from exothermic combustion reactions in the engine, +2 where fuel + air → $CO_2 + H_2O + \text{heat}$.

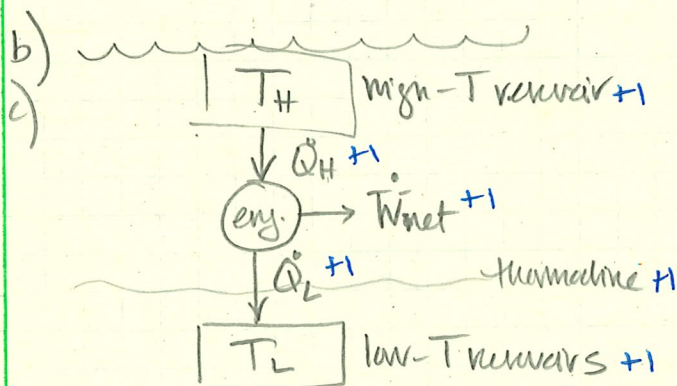
For a given $\dot{W}$, if $\eta_{th} \uparrow$, then $\dot{Q}_{in} \downarrow$, so less fuel is needed and $CO_2 \downarrow$. +3

[6] $T_H = 24^\circ C = 297.15 K$ +1

$T_L = 3^\circ C = 276.15 K$ +1

Assume: quasi-g., steady state
: max η_{th} means no losses

a) $\eta_{th, max} = 1 - \frac{T_L}{T_H} = 1 - \frac{276.15}{297.15} \rightarrow \boxed{\eta_{th, max} = 7.07\%}$ +1 not great...



* a lake is a reservoir because the engine operation would not change the entire temp of one of the Great Lakes. +2

d) Irreversibilities → friction

+2 for any

→ heat exchange b/w finite ΔT

→ turbulent mixing